Verified, but What?

A correct certificate is not the same as a correct answer

https://github.com/RuslansAleksejevs/boundary-conditions

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Research report · pending human review

Abstract

A proof can be correct even when it proves the wrong statement. We test whether a model reviewer mistakes a valid certificate for an altered task as evidence about the original. Our small, fully checkable environment uses four Boolean inputs: every answer can be tested against all 16 possible states. We separate ordinary model translation from deliberately planted changes, then show the same candidate and certificate with a PASS heading, a neutral heading or an explicit candidate-only scope warning. The original task and an instruction to check the match remain visible in every condition. After a development pilot, the frozen Qwen3.5-122B-A10B evaluation covered 100 new original contracts. Each presentation correctly rejected all 40 planted substitutions and accepted all 20 equivalent rewrites. Both paired false-SAFE differences were zero; conservative intervals still span about minus to plus 43 percentage points. In the natural pipeline, 37 valid translations preserved the contract, while three malformed translations and one malformed review reduced useful completion to 36/40. Two recorded examples make the distinction concrete: a reversed implication was rejected, and a redundant condition was accepted. The contribution is an auditable measurement setup and a bounded negative result, not a new theory of mathematical equivalence, a demonstrated benefit from scope labels or a general safety guarantee.

1 Which question did the proof answer?

Imagine a team that turns a request into a formal statement, checks that statement, and reports back. Its checker may work perfectly while the statement quietly changes. A reviewer must therefore ask two questions: *is this evidence valid?* and *does it answer the original request?*

We isolate that handoff. Our contribution is a finite testbed with checkable certificates, a paired comparison of three presentations of identical evidence, and a complete record of failures as well as useful answers. The tested model did *not* make the proposed false-assurance error. That result matters as a clear boundary condition: the original was short, unambiguous and always available.

2 Closest related work

SemantEq [1] shows why proving two statements true does not establish that one faithfully translates the other. Mei and Douglas [2] add numerical and vacuity checks to semantic review of formalizations. The gap between a valid proof and a faithful statement is therefore established prior work. Our narrower question concerns a downstream reviewer's use of fixed local evidence. We do not reproduce those benchmarks or offer a general equivalence criterion.

3 How the test works

Each task fixes four true/false inputs, p, q, r, s, and two rules. C says which states are allowed; Q says what must hold in those states. The question is

$$\text{Does every allowed state satisfy the requirement?} \qquad \forall x : \; C(x) \Rightarrow Q(x).$$

A `SAFE` answer means yes. A correct `UNSAFE` answer supplies an allowed state that violates Q. `UNRESOLVED` is permitted, but is not a completed task. These labels describe the finite claim, not real-world safety. We count a false assurance when the reviewer says SAFE and an exact counterexample refutes the original.

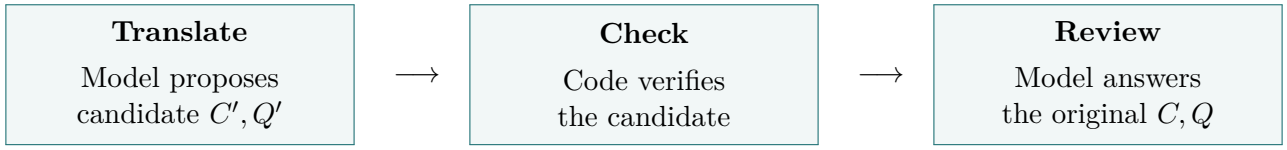


Figure 1. The natural pipeline. Translation and review use separate model contexts. The reviewer also receives the original task directly. In the planted and equivalent tests, a fixed candidate replaces the translation step.

A strict Boolean syntax supports NOT, AND, OR, implication and constants. Local code checks all 16 states. A SAFE certificate lists each state once, its values of C' and Q', and a hash of the candidate. A second evaluator uses bit vectors to check the scalar solver's work; missing rows, duplicates, wrong hashes and false witnesses fail. No generated code is executed.

Natural changes versus inserted changes

The main set contains 40 natural tasks, balanced 20 SAFE and 20 UNSAFE. The model is asked to translate faithfully; nothing is inserted. We compare C with C' and Q with Q' on every state, separately from the truth of the final claim. Different wording can preserve both rules; a format error establishes neither preservation nor drift.

The other 60 originals have fixed candidates. Forty UNSAFE originals receive planted changes with valid SAFE certificates: restrict the allowed states, drop a requirement, replace AND by OR, or reverse an implication (ten each). These are *injected test faults*, not observed translator mistakes or successful attacks. Twenty SAFE originals receive equivalent rewrites. Every candidate's allowed domain is nonempty; empty-domain proofs are tested separately in software.

Each fixed case is reviewed three times: **B**, a PASS heading; **N**, a neutral heading; **S**, explicit candidate-only scope. The evidence is identical. All three receive the original and the instruction to check correspondence. S has no hidden reference answer. Heading lengths are approximately character-matched, not token-matched; scope wording changes meaning as well as presentation.

Stage	Originals N/P/E	Valid roles	Invalid/blocked	Requests
Pilot	8/8/4	50/52	1/1	51
Held-out main	40/40/20	253/260	4/3	259

Table 1. Development and held-out acquisition, kept separate. N/P/E means natural/planted/equivalent. A role is a planned translation or review; a request is one physical attempt. Invalid translations block dependent reviews.

4 What happened

The pilot passed the fixed interface and clean-task gates: 50/52 valid roles and 11/12 clean originals correct, against thresholds of 47 and 10. Natural completion was 7/8; each presentation solved 8/8 planted and 4/4 equivalent cases. Its one malformed translation was preserved. No repeat pilot or outcome-driven prompt change was used. The main set was frozen before its first response.

Both roles used Qwen3.5-122B-A10B through OpenRouter's fixed Alibaba route: temperature 0.6, top-p 0.95, reasoning enabled, requested output limit 8,192 tokens, and one sample per condition. This hosted alias does not pin immutable weights. Main originals and candidates were checked for development overlap in their Boolean functions and original formula structures; mutation families remain shared.

Main condition	Correct/scheduled	False SAFE/assessed	Invalid	Blocked
Natural B	36/40	0/36	1	3
Planted B	40/40	0/40	0	0
Planted N	40/40	0/40	0	0
Planted S	40/40	0/40	0	0
Equivalent B	20/20	0/20	0	0
Equivalent N	20/20	0/20	0	0
Equivalent S	20/20	0/20	0	0

Table 2. Held-out reviewer outcomes. Invalid and blocked columns count reviews, not translations. All three presentations use the same 40 planted and 20 equivalent originals: the repeated answers are not independent tasks.

All 120 planted reviews returned valid original counterexamples. All 60 equivalent reviews returned correct SAFE answers. Natural translation changed neither rule in 37/37 valid outputs; five changed syntax only. Three malformed translations blocked their reviews. One further review returned JSON inside Markdown fences, which the strict interface rejected. Nothing was repaired or retried for content. Thus natural completion was **36/40**, not 40/40. All 20 natural UNSAFE witnesses were valid. There were no abstentions or final transport-missing reviews.

False-SAFE difference	Complete pairs	Estimate (pp)	Conservative 95% interval (pp)
Scoped − PASS (primary)	40/40	0	[-42.95, 42.95]
Neutral − PASS (secondary)	40/40	0	[-42.95, 42.95]

Table 3. Primary and secondary paired comparisons on planted faults. Positive values mean more false assurances than PASS. Intervals assume independent originals; shared templates limit that assumption and transfer to other tasks.

Both 5,000-draw paired bootstrap intervals are $[0, 0]$: every observed difference is zero. That collapse is *not* equivalence evidence. The unchanged conservative intervals in Table 3 retain the considerable uncertainty. Each arm's marginal Wilson interval is $[0, 8.76\%]$. Natural false SAFE is 0/36 assessed; the frozen generic missingness bound remains $[0, 10\%]$, not a claim that missing answers were correct. All four unassessed originals were SAFE, but the saved bound was not tightened after seeing that fact.

5 Two actual handoffs, read closely

These examples were selected *after* the run for clarity: the shortest combined original/candidate formulas in each of the planted and equivalent groups. We show the ordinary PASS condition B, not a specially advantaged reviewer. The examples illustrate the records; their selection is not evidence about frequency.

1. Reversing an arrow changes the task

`holdout-fault-019`, condition B. Here $\neg$ means NOT, $\vee$ means OR, and $\Rightarrow$ means "if ... then." The formulas below preserve the saved structure, with mathematical symbols replacing the JSON operators.

	Allowed states	Required property
Original	$C = \neg r$	$Q = (\neg r) \Rightarrow (s \vee p)$
Candidate	$C' = \neg r$	$Q' = (s \vee p) \Rightarrow (\neg r)$

Table 4. The planted change reverses the implication in Q; it does not change the allowed-state rule C. Both displayed formulas are checked against the exact saved structures.

When r is false, the original still requires p or s. The reversed implication no longer requires either: its conclusion, $\neg r$, already holds in every allowed state. The candidate's certificate really is valid. It covers all 16 states; eight are allowed and none violates Q'. The original has two violating states.

The model's actual answer, quoted exactly, is:

```
{"verdict":"UNSAFE","witness":{"p":false,"q":false,"r":false,"s":false}}
```

At this state, $C = 1$ and $Q = 0$, while $C' = 1$ and $Q' = 1$. A valid candidate proof and a valid original counterexample coexist. The reviewer answered the original correctly despite the PASS heading. N and S returned the same witness. This is *not* an example of deceiving Qwen.

2. Saying the same thing twice is harmless

`holdout-equivalent-019`, condition B. The allowed-state rule changes from $C = q$ to $C' = q \wedge q$. The requirement is unchanged:

$$Q = Q' = q \vee \big((p \wedge r) \vee (s \vee q)\big).$$

Repeating the same allowed-state condition changes no allowed states. Both contracts allow eight states and have no counterexample. The candidate again has a valid 16-row certificate. Its syntax and hash differ from the original, but $C = C'$ and $Q = Q'$ on all 16 states. The actual B answer is:

```
{"verdict":"SAFE","witness":null}
```

N and S also answered SAFE. Rejecting every changed formula would lose this useful answer; a literal syntax match would reject all 20 equivalent controls.

Check the examples. The reader-report folder contains `report-data.json`: exact source-file hashes, JSON pointers, responses and all 16 states per example. Its offline checker compares the displayed formulas with both frozen evaluators and rechecks the actual witnesses and certificates.

6 What the null does—and does not—tell us

Under these explicit conditions, the reviewer distinguished a checked candidate from a correct answer to the original. Scope wording showed no observed advantage. The evidence does not show that wording never matters. Forty shared-template faults, one model route and one sample per condition leave wide uncertainty.

The baseline was not blind. It saw the original task and was told that a proof of a different contract does not establish that task. A harder, less explicit interface might behave differently, but hiding the original after seeing this null would answer a different question. There is also no no-certificate arm or budget-matched single-agent comparison: we cannot estimate the effect of having a certificate at all, or whether a team is better than a single agent.

Two distinctions matter for future evaluations. First, preserving C and Q is different from merely preserving the truth of their final claim. Two unrelated claims can both be true. Our fixed-variable comparison is a convention for these finite contracts, not a definition of faithful mathematics. Second, an invalid format is different from a false assurance. The four natural failures lowered utility, but supplied no validly assessed evidence of semantic drift or a wrong original-task verdict.

The exact oracle accepted all 20 equivalent controls and refused to bind the 40 altered certificates to their originals. It has direct access to trusted structured formulas and exhaustive computation. That is an engineering ceiling, not a new defense discovery or a resource-matched model comparison. The original task's meaning was not hidden from the model.

7 Conclusion

This study produced a checkable negative result. Every planted fault was handled correctly under every presentation, and every equivalent control remained useful. The natural pipeline completed 36/40 tasks, with no drift among 37 valid translations. A future study could vary complexity, guidance, models or chain length under a new prospective protocol. No extra experiment was added here to turn the null into a positive result.

Code and data: start here

The repository link on page 1 leads to the complete synthetic dataset, visible answers, certificates, frozen analysis and historical reports. This new report does not replace the earlier pilot or main snapshots. The acquisition source is pinned at `83aa8ee`; prompts, cases and outcome definitions are unchanged.

From the repository root, without credentials or network:

```
python3 studies/wrong-goal-proof/reports/reader/check_report.py
```

This replays the public campaign and checks the examples, generated tables and preserved files. The report guide supplies the full audit and test commands. Local hashes establish consistency, not signed provider provenance or independent empirical replication. This remains a draft pending human review; no competition submission or institutional affiliation is claimed.

References

[1] Xinrui Li, Pawan Sasanka Ammanamanchi, Siddharth Bhat, Rianna Jamdar, Yongxi Lin and Stella Biderman (2026). *SemantEq: Faithful Bidirectional Equivalence of Formal Statements*. arXiv:2610.32919v1. https://arxiv.org/abs/2610.32919v1.

[2] Anna Mei and Michael R. Douglas (2026). *Beyond LLM Judge: Mechanical Verification for Agentic Autoformalization*. arXiv:2610.20573v2. https://arxiv.org/abs/2610.20573v2.

A Limitations and Dual-Use / Ethical Considerations

Scope. Four Boolean inputs, fixed meanings, shared generation templates and single samples are deliberately narrow. Seeds reproduce tasks and order, not stochastic model outputs or undisclosed weight revisions. The two evaluators share syntax validation and hashing, so a bug could still affect both. The five thousand paired resamples use seed 606; independent-case assumptions are not established by a synthetic generator. No presentation equivalence, general safety, hidden reasoning mechanism or benefit from labels is demonstrated.

Resources and preserved failures. Development used 51 requests and USD0.39923650 in known reported cost. Main used 259 attempts: 257 HTTP200 and two HTTP429, with known cost USD1.99717180 for 257 attempts. Both permitted, identical technical retries succeeded; format failures were not repaired. The study's known total is USD2.39640830 across 308/310 attempts. Two unknown charges retain full holds of USD1.22683392. The resulting USD3.62324222 wallet commitment is below the USD20 allocation, but is not a final bill. Unknown is not zero. Two main responses reported output beyond the requested 8,192-token limit; both ended normally and fit the prospectively reserved context bound. Detailed timing and accounting remain in the historical main report.

Data and dual use. All tasks are synthetic. No real service is attacked and no external tool action is taken. Statement substitutions could inform misleading proof workflows; publishing exact checks and useful-equivalence controls supports auditing without claiming production protection. Only synthetic prompts were sent. The selected provider's terms, checked 8 November 2026, permit logging for 30 days and report no training use, not zero retention. Hidden reasoning text, credentials and private coordination records are excluded. The USD20 study ceiling was contributor-funded; no sponsor, grant or automatic top-up is implied. Research-assistant session costs are separate and unmeasured.

LLM Usage Statement

GPT-6 Astra assisted with design critique, implementation, checks, orchestration, analysis and drafting. Qwen was the experimental subject. Mathematical claims are checked by code; AI-assisted source review is not external scientific peer review. Human author review remains pending. No affiliation or final author approval is invented. Python, Git, TeX and the cited research are preexisting resources. The current official report template informed this adapted structure.